Chapter 20 worked solutions • Influential operating systems

OS20-E01 • Spool bottleneck

1. CPU J1 0–2, J2 2–4, J3 4–6; printer J1 2–6, J2 6–10, J3 10–14. Output waits without interrupting the printer's current job.
2. Total 14, CPU 6/14=3/7, printer 12/14=6/7. Serial execution takes 3(2+4)=18; CPU utilization is 6/18 = 1/3 and printer utilization is 12/18 = 2/3. One printer still performs 12 rather than 4 units of service.
3. Spooling retains jobs/data/queues to decouple producers/consumers. Multiprogramming schedules other ready CPU work while one waits. Spools alone supply neither multiple resident jobs nor scheduling/isolation. Full queues/timing substantiate utilization claims.

OS20-E02 • THE and Mach boundaries

1. Both decompose services/responsibilities. THE emphasizes abstraction/synchronization; Mach 3 user servers additionally use protected address spaces/IPC. Modularity alone does not isolate faults.
2. The specified separate user-server IPC entails crossings. A THE layer call need not trap; Mach 2.x retains much BSD code in kernel and differs.
3. Establish C0; each Ci assumes lower contracts/stated interference, establishes itself, and preserves lower hidden state. Induction then applies. Cycles or direct uncontrolled lower-state changes violate premises. Layer, ring, process differ.

OS20-E03 • Atlas branches and faults

1. All-zero set uses branch 3, maximum T = 5 selects B. Other set has no t>T+1; branch 2 ignores C with t = 0 and compares scores A = 2 and B = 5, selecting B. Omitting branch 3 leaves a legal state undefined.
2. 3597=7 • 512+13→2 • 512+13=1037 physical words, not bytes.
3. Both require valid mapping, correct contents, restartable faults, frame ownership. Modern multilevel tables, TLBs, COW, sharing, permissions, multicore synchronization add conditions. Atlas predictions do not guarantee the minimum fault count achieved by OPT.

OS20-E04 • XDS relocation/sharing

1. Eight virtual pages, 32 frames. 6161=3 • 2048+17→11 • 2048+17=22545 words.
2. Two shared+two • six private=14, not 16 or 12. The question specifies identical physical frames for the two shared code pages and that all pages are resident. Code must be safely read-only or otherwise safe to share; mutable data remains separate. Multiple mappings do not privatize shared bytes.
3. No. This case relocates resident pages. Demand loading, restartable missing-page faults, and swapping policy are additional mechanisms. Separate units from residency policy.

OS20-E05 • Two-buffer conservation

1. After sends, free∅, queued{b0, b1}. After reply, free∅, queued{b1}, reply{b0}, total 2. If processing not yet answered, place b0 only in processing instead.
2. No while reply owns it. After C1 consumes/returns, b0 becomes free and C3 can claim it. Each buffer has one state.
3. No. C1 awaiting S while S awaits C1 forms circular wait. Primitive atomicity protects its transition, not whole transactions/system progress. Conservation and wait relationships need separate checks.

OS20-E06 • Exponential feedback

1. Allowed 1/2/4/8, used 1/2/4/4, remaining 10/8/4/0: four dispatches. The final slice ends early on completion.
2. Feedback 11 + 4 • 0.5 = 13; RR 11 + 11 • 0.5 = 16.5. These are model costs, not historical speed measurements.
3. Full service through k sums 1+2+…+2^k=2^(k+1)−1. Blocking, promotion, competitors change burst/wait traces. Larger quanta alone do not prevent low-priority starvation. Do not mix different feedback rules.

OS20-E07 • Multics segments/pages

1. 1024=2^10, so the 18-bit offset splits into 8-bit page/10-bit displacement. 2500=2 • 1024+452; PA = 7 • 1024+452=7620 words, after descriptor bound/rights checks.
2. Rings enforce privilege and entry boundaries, while layers organize dependencies and abstractions. On GE-645, ring crossings used supervisor traps and software; hardware ring support came with the later Honeywell 6180. Modules can share a ring; six layers do not prove six domains.
3. Resolution identifies/references an object. Access lists, operation, execution context still authorize it. Shared writable segments still need synchronization. Mapping and authorization are separate steps.

OS20-E08 • MFT/MVT and base-limit

1. Two 50 KiB jobs consume 64 KiB each, wasting 28 KiB internally. The remaining 128 KiB cannot fit 70 KiB in one region. Variable holes total 120 KiB but the maximum hole of 50 KiB cannot fit 70 KiB, external fragmentation.
2. Validate $0 \le a < 768$. Then $0 \to 256$ and $767 \to 1023$; 768 and −1 fail. Protected setup is essential; user-modifiable bounds destroy confinement.
3. CP supplies VM resource isolation; CMS runs as a single-user guest. Multiple guests do not make CMS one shared-memory multiuser kernel. Configuration, relocation, VM abstraction differ. Total free space does not replace a suitable interval.

OS20-E09 • Specified Mach 3 copying

1. The request path C user → C kernel → S kernel → S user makes 2 mode changes and 1 address-space change, copying 64 B. The reply follows the reverse path and makes 2 mode changes and 1 address-space change, copying 8192 B. Total: 4 mode switches, 2 address-space switches, and 8256 B copied. Extra interrupts/scheduling are excluded; changed premises require recalculation.
2. Protection limits server directly corrupting client/kernel memory. Clients still handle timeout, failed state, reconnection, retry. Neither universal availability nor bug-free kernel follows.
3. Mach 2.x retains much BSD in kernel; derivatives can reorganize services. Version and actual placement, not a Mach name, determine IPC costs. Mode, address-space, and byte counts differ.

OS20-E10 • Names versus mechanisms

1. Cooperative yield/block-only switching may starve B forever. Preemptive progress requires serviceable timer, bounded kernel blocking, and scheduling that gives B time. Two windows prove neither processes nor protection.
2. No. Here completion only supplies the command name; it does not mean a write system call succeeded. Syntax assistance creates a recognizable request; trusted identity, target, right still determine access. Names can also change between checking/opening.
3. No. Languages improve expression/checks; small kernels can narrow trusted scope. Synchronization, ownership, authority, and failure contracts still need proof. Usability, isolation, progress, and overall correctness are distinct.

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz et al., Operating System Concepts, 9th ed., Ch. 20. Atlas one-level store; THE system; Multics GE-645; Multics protection.

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https://github.com/xinhi6716/computer-science-self-study